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EXPERIMENT-11

Question

Write a **menu-driven C program using switch-case** to input a positive integer and perform one of the following operations:

1. Find the **sum and product of the first and last digits**.
2. Count how many digits are **lesser than, equal to, and greater than** a given digit k .

1. Algorithm

Step 1:Start

Step 2:Input a positive integer n

Step 3:Display the menu:

1. Sum and product of first and last digits
2. Count digits lesser than, equal to, and greater than k

Step 4:Input the user's choice

Step 5:Use a `switch` statement according to the choice

Step 6: If choice = 1

- Store the last digit using `last = n % 10`
- Copy n into another variable `temp`
- Divide `temp` by 10 repeatedly until it becomes less than 10
- The remaining value is the first digit
- Calculate `sum = first + last`
- Calculate `product = first * last`
- Display the sum and product

Step 7: If choice = 2

- Input digit k

- Set `lesser = 0`, `equal = 0`, and `greater = 0`
- Copy `n` into `temp`
- Extract each digit using `temp % 10`
- If the digit is less than `k`, increase `lesser`
- If the digit is equal to `k`, increase `equal`
- Otherwise, increase `greater`
- Divide `temp` by 10 and repeat until all digits are checked
- Display the three counts

Step 8: If the choice is neither 1 nor 2, display **Invalid choice**

Step 9: Stop

2. Test Case Table

Test Case	Inputs	Expected Output	Actual Output	Results
1	n = 12345, Choice = 1	Sum = 6, Product = 5	Sum = 6, Product = 5	Pass
2	n = 8246, Choice = 1	Sum = 14, Product = 48	Sum = 14, Product = 48	Pass
3	n = 12345, Choice = 2, k=3	Lesser = 2, Equal = 1, Greater = 2	Lesser = 2, Equal = 1, Greater = 2	Pass
4	n = 55525, Choice = 2, k=5	Lesser = 1, Equal = 4, Greater = 0	Lesser = 1, Equal = 4, Greater = 0	Pass
5	n = 9876, Choice = 2, k=7	Lesser = 1, Equal = 1, Greater = 2	Lesser = 1, Equal = 1, Greater = 2	Pass

3. C Program

```
#include <stdio.h>
int main()
{
    int n, choice, temp, first, last;
    int sum, product;
    int k, digit;
    int lesser = 0, equal = 0, greater = 0;
    printf("Enter a positive integer: ");
    scanf("%d", &n);
    printf("\n1. Find sum and product of first and last digits");
    printf("\n2. Count digits lesser than, equal to and greater than k");
    printf("\nEnter your choice: ");
    scanf("%d", &choice);
    switch(choice)
    {
        case 1:
            last = n % 10;
            temp = n;
```

```

while(temp >= 10)
{
    temp = temp / 10;
}
first = temp;
sum = first + last;
product = first * last;
printf("Sum = %d\n", sum);
printf("Product = %d\n", product);
break;
case 2:
    printf("Enter digit k: ");
    scanf("%d", &k);
    temp = n;
    while(temp > 0)
    {
        digit = temp % 10;
        if(digit < k)
            lesser++;
        else if(digit == k)
            equal++;
        else
            greater++;
        temp = temp / 10;
    }

    printf("Count_lesser = %d\n", lesser);
    printf("Count_equal = %d\n", equal);
    printf("Count_greater = %d\n", greater);
    break;
default:
    printf("Invalid choice\n");
}
return 0;
}

```

4. Compilation and Execution

```

(kali㉿kali)-[~]
$ gcc hdoc12.c -o hdoc12

(kali㉿kali)-[~]
$ ./hdoc12
Enter a positive integer: 12345

1. Find sum and product of first and last digits
2. Count digits lesser than, equal to and greater than k
Enter your choice: 1
Sum = 6
Product = 5

(kali㉿kali)-[~]
$ gcc hdoc12.c -o hdoc12

(kali㉿kali)-[~]
$ ./hdoc12
Enter a positive integer: 8246

1. Find sum and product of first and last digits
2. Count digits lesser than, equal to and greater than k
Enter your choice: 1

```

```

Sum = 14
Product = 48

(kali㉿kali)-[~]
$ gcc hdoc12.c -o hdoc12

(kali㉿kali)-[~]
$ ./hdoc12
Enter a positive integer: 12345

1. Find sum and product of first and last digits
2. Count digits lesser than, equal to and greater than k
Enter your choice: 2
Enter digit k: 3
Count_lesser = 2
Count_equal = 1
Count_greater = 2

(kali㉿kali)-[~]
$ gcc hdoc12.c -o hdoc12

(kali㉿kali)-[~]
$ ./hdoc12
Enter a positive integer: 55525

1. Find sum and product of first and last digits
2. Count digits lesser than, equal to and greater than k
Enter your choice: 2
Enter digit k: 5

```

```

Count_lesser = 1
Count_equal = 4
Count_greater = 0

(kali㉿kali)-[~]
$ gcc hdoc12.c -o hdoc12

(kali㉿kali)-[~]
$ ./hdoc12
Enter a positive integer: 9876

1. Find sum and product of first and last digits
2. Count digits lesser than, equal to and greater than k
Enter your choice: 2
Enter digit k: 7
Count_lesser = 1
Count_equal = 1
Count_greater = 2

```

5. Testing

The program was compiled and executed for the prepared test cases. The actual outputs matched the expected outputs for all the test cases. Therefore, all test cases passed successfully.